1 Model

We consider a chain of N qubits evolving under a noisy circuit. The reduced density matrix obeys a master equation whose generator we write down explicitly in what follows, as in many earlier works on the subject of open quantum systems.

$$\frac{d\rho}{dt} = -i[H,\rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right) \tag{1}$$

Here H is the Hamiltonian and L_k are jump operators. The rates γ_k are positive, which guarantees complete positivity of the map generated by Eq. (1). We now expand the Hamiltonian in Pauli strings, see (1).

$$H = \sum_j J_j Z_j Z_{j+1} + h \sum_j X_j, \tag{2}$$

$$E = \langle \psi | H | \psi \rangle. \tag{3}$$

The energy in Eq. (3) is bounded. We also need the entropy of a subsystem, defined by

$$S_A = -\mathrm{Tr}\, \rho_A \log \rho_A. \tag{4}$$

A long paragraph follows to fill the column with prose text so that equations appear in both columns of the page and the detection has to deal with the gutter between the columns properly and not mix them up at all. More words are added here so that the text is long enough.

$$F(t) = \int_0^t ds\, e^{-\lambda(t-s)} \frac{\sin \omega s}{s} \tag{5}$$

Equation (5) closes the derivation, and with it the section on the model, which we summarise in a figure below.

$$x = y \quad \text{(unnumbered)}$$



Figure 1: The phase diagram of the model.

More text at the end of the second column, describing results with $x = 1$ and nothing else in particular.

$$\begin{aligned} G = a + b + c + d + e + f \\ + g + h + i + j \end{aligned} \tag{6}$$

2 Intro

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$$a^2 + b^2 = c^2. \tag{2.1}$$

We reference (1.1) at the start of nothing. A very wide equation follows:

$$Z = \sum_{\sigma_1, \sigma_2, \ldots, \sigma_N} \exp\left(\beta \sum_{i<j} J_{ij}\sigma_i\sigma_j + \beta h \sum_i \sigma_i + \beta g \sum_{i<j<k} K_{ijk}\sigma_i\sigma_j\sigma_k + \beta\lambda\right) \tag{2.2}$$

$$\dot{x} = -\gamma x + \sqrt{2D}\,\xi(t), \tag{2.3a}$$

$$\dot{y} = x - y, \tag{2.3b}$$

where ξ is white noise, as stated in (1.3a) and (1.3b).

$$\mathcal{L}[\rho] = \sum_k \left(L_k \rho L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho\}\right) \tag{2.4}$$

$$M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix} \tag{2.5}$$

Final text of the section, (1.5) refers back

A Details

$$\int_0^\infty e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2} \tag{A1}$$